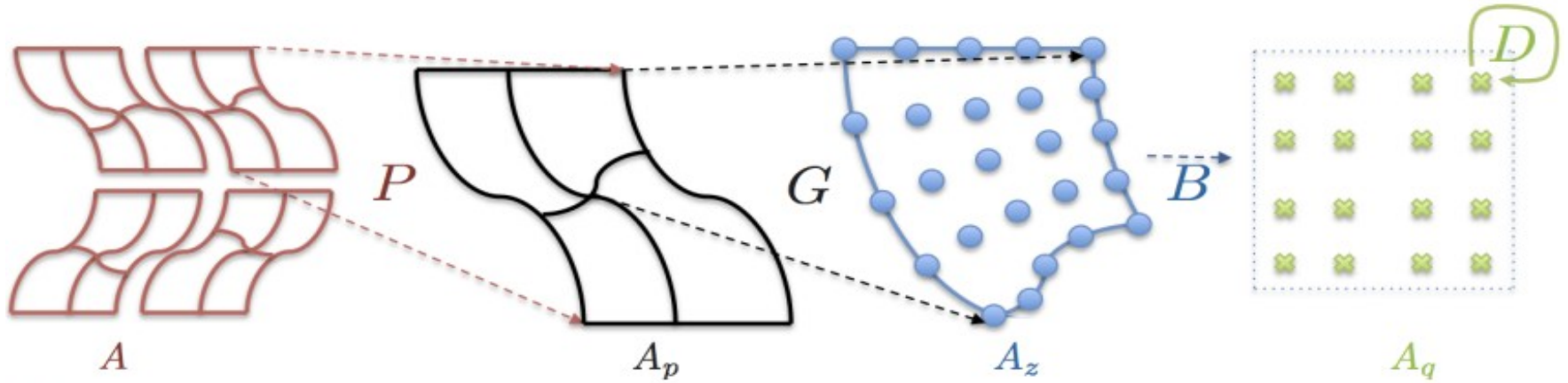
Low-level API (libCEED)

$$A = P^T G^T B^T D B G P$$

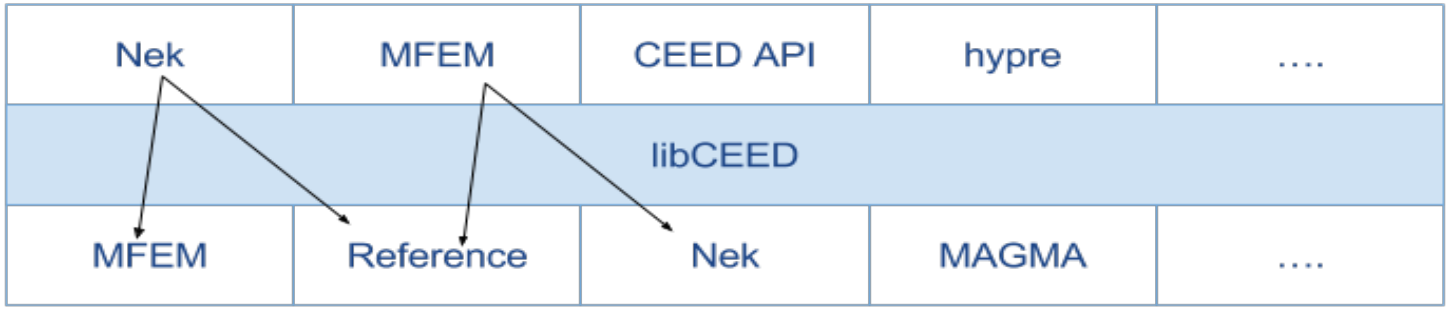


Parallel, Mesh Topology, Finite Element Basis, Geometry/Physics

- A suite of higher order operator discretization that enables efficient evaluation using BP winners as back-end evaluation kernels.
- For each operator we have:
  - Data structure representing the assembled operator
  - Operator assembly routine
  - Operator action routine
- Building blocks for the high level API:
  - Multiple element types, e.g. quad+tri
  - Variable polynomial orders
- Use *efficient* operator representations

```
void diffusion_action
(
   double *y,                 /* result, input-output vector */
   double *x,                 /* input vector */
   double *D,                 /* nqpt_1d x nqpt_1d x nelem */
   double *B1d,               /* nqpt_1d x ndof_1d dense matrix */
   double *B1d_t,             /* trasnspose of B1d */
   double *G1d,               /* nqpt_1d x ndof_1d dense matrix */
   double *G1d_t,             /* trasnspose of G1d */
   int *dof_offsets,          /* array of size
                                 (ndofs_1d ^ dim) x nelem
                                 representing a boolean P */
   int ndof_1d,               /* number of 1D dofs (points) */
   int nqpt_1d,               /* number of 1D quadrature points */
   int nelem,                 /* number of elements */
   int ndim,                  /* Dimension of the problem */
   int backend,               /* Backend type */
   int vtype,                 /* Vector type */
   int olayout                /* Operator type */
);
```

Low-level API (libCEED)



Current Status

- Reference, Nek5000 and MFEM backend implementations (More optimizations to be done).
- MFEM can call Reference and Nek5000 back-ends for Mass and Diffusion actions.
- Working on using MFEM and Reference back-ends from Nek5000

Future Plans

- Figure out the correct API that should be exposed to the high-level user applications.
- Implement more actions, combined actions, etc.
- Expand and optimize for various devices, GPUs, KNLs, etc.